

AKSHATHA RAI K

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Kasaragod, Kerala, India

Career Objective

Motivated MCA student with hands-on experience in Full Stack development and AI-based applications. Seeking an opportunity to begin a career in software development with a strong interest in learning and building expertise in SAP technologies, Java, React and cloud-based digital solutions while contributing to real-world projects and continuously developing technical skills.

Education

Master of Computer Application	Present
NMAM Institute of Technology, Nitte	CGPA: 8.46
Bachelor of Science	2025
Vivekananda College of Arts, Science and Commerce (Autonomous), Puttur	CGPA: 8.6
PUC (12th)	2022
Sri Rama PU College, Kalladka	83.5%
SSLC (10th)	2020
Sri Satya Sai Loka Seva High School, Alike	88%

Technical Skills

Languages: C, Java, JavaScript, Python
Frontend: HTML, CSS, Bootstrap, React.js
Backend: Node.js, Express.js, Flask
Database: MySQL, MongoDB (Basic)
AI/ML: PyTorch, TensorFlow/Keras, OpenCV, NumPy, CNN
Tools: Git, GitHub, VS Code, Microsoft Office
Concepts: OOP, SDLC, Operating Systems, REST APIs

Projects

AadhaarVoteGuard: Biometric Voting Security System

- Built RESTful APIs using Node.js and Express.js with JWT authentication and bcryptjs password hashing.
- Integrated Nodemailer for OTP delivery with 5-minute expiry and MySQL transactions to prevent duplicate voting.
- Developed a React.js frontend using React Router and Axios for API communication.
- Followed separated frontend-backend architecture and managed source code using Git.

Real-Time Low-Light Obstacle Identification for Visually Impaired Persons Using CLAHE-Enhanced CNN on Smartphone Camera with Audio Feedback

- Developing an AI-based obstacle detection system using a smartphone camera to assist visually impaired users.
- Applying CLAHE-based low-light enhancement and CNN-based real-time obstacle detection and classification.
- Implementing audio feedback to alert users about detected obstacles for safer navigation.
- Technologies:** Python, OpenCV, TensorFlow/Keras, CNN, CLAHE.

Handwritten Signature Verification using Siamese Neural Network

- Developing a deep learning-based system to verify handwritten signatures by comparing reference and test images.
- Implemented image preprocessing using OpenCV, Pillow, and NumPy, including grayscale conversion, resizing, and normalization.
- Implemented a Siamese CNN using PyTorch to extract signature features and determine similarity.
- Built a Flask-based web application and REST API to classify signatures as genuine or forged.
- Technologies:** Python, PyTorch, Flask, OpenCV, NumPy, CNN, Siamese Neural Network.

Certifications

- Cloud Computing and Distributed Systems – NPTEL SWAYAM April 2026
- Oracle Cloud Infrastructure 2025 Certified Foundations Associate – Oracle University May 2026
- Microsoft AINNOVATION 2025: Applied AI Learning Challenge and Microsoft Azure Learning Challenge August 2025